9 Illustrate the concept of inter-process communication using shared memory with a C program

main.c

Run ▶

```c
#include <stdio.h>
#include <stdlib.h>
#include <sys/ipc.h>
#include <sys/shm.h>
#include <sys/wait.h>
#include <unistd.h>
#include <string.h>
int main() {
    int shmid;
    char *shared_memory;
    shmid = shmget(IPC_PRIVATE, 1024,
                   IPC_CREAT | 0666);
    if (shmid < 0) {
        perror("shmget failed");
        return 1;
    }
    shared_memory = (char *)shmat(shmid, NULL, 0);
    if (shared_memory == (char *)-1) {
        perror("shmat failed");
        shmctl(shmid, IPC_RMID, NULL);
        return 1;
    }
    pid_t pid = fork();
    if (pid < 0) {
        perror("fork failed");
        shmdt(shared_memory);
    }
    if (pid == 0) {
        strcpy(shared_memory,
               "Hello from child process!");
        printf("Child: Message written to shared
memory.\n");
    } else {
        wait(NULL);
        printf("Parent: Message received: %s\n",
               shared_memory);
    }
    shmdt(shared_memory);
    if (pid > 0) {
        shmctl(shmid, IPC_RMID, NULL);
    }
    return 0;
}
```

Output

```
Child: Message written to shared memory.
Parent: Message received: Hello from child process!
```